Volume: Mixed Practice

Name _____ Date _____ Calculator allowed Version 1

Scan to practise this topic online

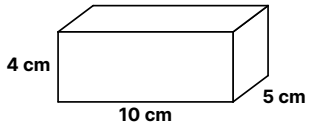
KEY WORDS volume *(rōrahi)* · prism *(poro)* · cross-section *(hautanga whakapae)* · cylinder *(porotaka)*

A Rectangular prisms ●○○

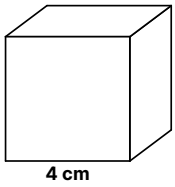
Remember: Volume of a rectangular prism = length × width × height. Volume is measured in cubed units, such as cm^3.

1. Find the volume of each solid.

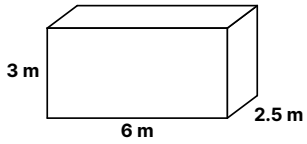
a)



b)



c)



2. Find the volume of:

a) A rectangular prism 12 cm by 8 cm by 5 cm

b) A cube with edges of 9 mm

c) A rectangular prism 2.5 m by 1.2 m by 0.8 m

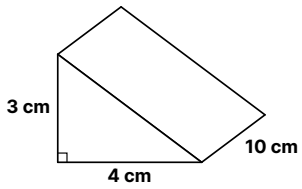
d) A rectangular prism 20 cm by 15 cm by 6 cm

B Other prisms ●●○

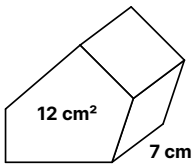
Remember: Volume of a prism = area of the end face × length.

3. Find the volume of each prism.

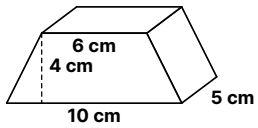
a)



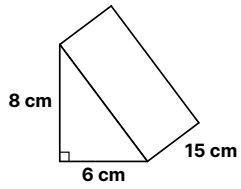
b)



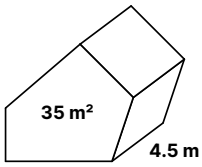
c)



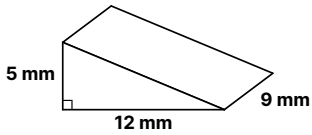
d)



e)



f)



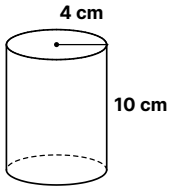
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C Cylinders ●●○

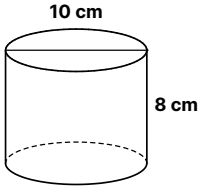
Remember: Volume of a cylinder $= \pi r^2 h$. Round to 1 decimal place.

4. Find the volume of each cylinder.

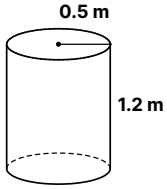
a)



b)



c)



5. Find the volume of:

a) A cylinder with radius 6 cm and height 15 cm

b) A cylinder with diameter 7 cm and height 12 cm

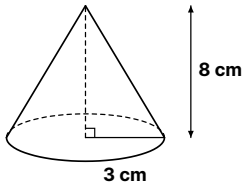
c) A cylinder with radius 1.5 m and height 4 m

D Cones, pyramids and spheres ●●●

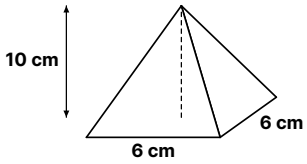
Remember: Cone or pyramid: $V = \frac{1}{3} \times \text{base area} \times \text{height}$. Sphere: $V = \frac{4}{3}\pi r^3$.

6. Find the volume of each solid. Round to 1 decimal place where needed.

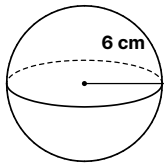
a)



b)



c)



7. Find the volume of:

a) A cone with radius 6 cm and height 14 cm

b) A sphere with radius 2.1 m

c) A hemisphere (half a sphere) with radius 9 cm

E Problems ●●●

8. A fish tank is 60 cm long, 30 cm wide and 40 cm high. How many litres of water does it hold when full? (1 litre = 1000 cm³)

9. A water tank is a cylinder with a radius of 1.5 m and a height of 2 m.

a) Find its volume in m³, to 1 decimal place. **b)** How many litres is this? (1 m³ = 1000 L)

10. A box is 40 cm by 30 cm by 20 cm. It is packed full of cubes with edges of 5 cm. How many cubes fit in the box?

11. A prism has a volume of 360 cm³. The area of its end face is 24 cm². Find the length of the prism.

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Extension

12. A cylinder has a volume of 500 cm^3 and a radius of 4 cm. Find its height, to 1 decimal place.

13. A metal sphere with a radius of 3 cm is melted down and made into a cylinder with a radius of 2 cm. Find the height of the cylinder.

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Volume: Mixed Practice

Name _____ Date _____ Calculator allowed Version 2

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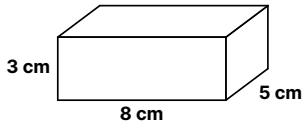
KEY WORDS volume *(rōrahi)* · prism *(poro)* · cross-section *(hautanga whakapae)* · cylinder *(porotaka)*

A Rectangular prisms ●○○

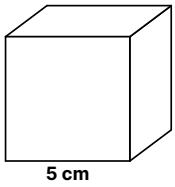
Remember: Volume of a rectangular prism = length × width × height. Volume is measured in cubed units, such as cm^3.

1. Find the volume of each solid.

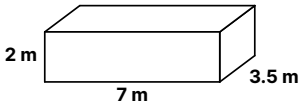
a)



b)



c)



2. Find the volume of:

a) A rectangular prism 15 cm by 6 cm by 4 cm

b) A cube with edges of 7 mm

c) A rectangular prism 3.5 m by 1.4 m by 0.6 m

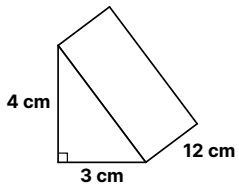
d) A rectangular prism 25 cm by 12 cm by 8 cm

B Other prisms ●●○

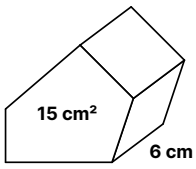
Remember: Volume of a prism = area of the end face × length.

3. Find the volume of each prism.

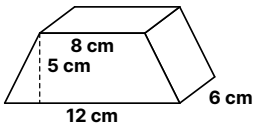
a)



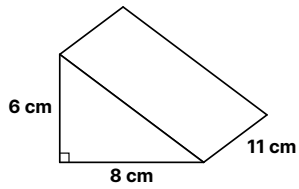
b)



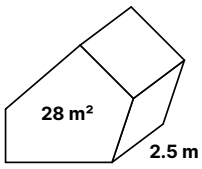
c)



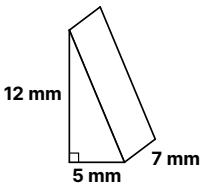
d)



e)



f)



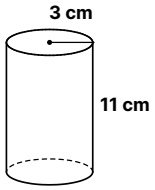
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C Cylinders ●●○

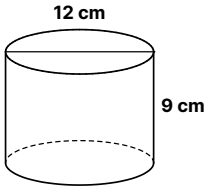
Remember: Volume of a cylinder $= \pi r^2 h$. Round to 1 decimal place.

4. Find the volume of each cylinder.

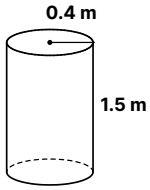
a)



b)



c)



5. Find the volume of:

a) A cylinder with radius 5 cm and height 18 cm

b) A cylinder with diameter 9 cm and height 14 cm

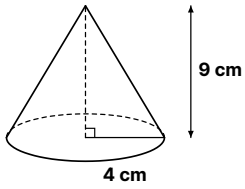
c) A cylinder with radius 2.5 m and height 3 m

D Cones, pyramids and spheres ●●●

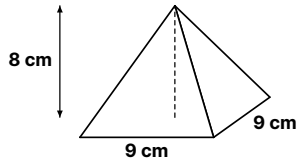
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6. Find the volume of each solid. Round to 1 decimal place where needed.

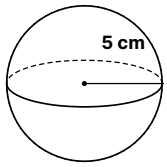
a)



b)



c)



7. Find the volume of:

a) A cone with radius 5 cm and height 12 cm

b) A sphere with radius 1.8 m

c) A hemisphere (half a sphere) with radius 6 cm

E Problems ●●●

8. A fish tank is 50 cm long, 40 cm wide and 30 cm high. How many litres of water does it hold when full? (1 litre = 1000 cm³)

9. A water tank is a cylinder with a radius of 2 m and a height of 1.5 m.

a) Find its volume in m³, to 1 decimal place. **b)** How many litres is this? (1 m³ = 1000 L)

10. A box is 60 cm by 40 cm by 20 cm. It is packed full of cubes with edges of 10 cm. How many cubes fit in the box?

11. A prism has a volume of 450 cm³. The area of its end face is 30 cm². Find the length of the prism.

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Extension

12. A cylinder has a volume of 800 cm^3 and a radius of 5 cm. Find its height, to 1 decimal place.

13. A metal sphere with a radius of 6 cm is melted down and made into a cylinder with a radius of 4 cm. Find the height of the cylinder.

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